

ISC Class XII Physics

Chapter 12 Atoms

30 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. α -Scattering Experiments
- II. Radii of Orbits and Velocities in these Orbits
- III. Energy of Electron in Different Orbits
- IV. Energy Emitted/Absorbed in Transitions
- V. Excitation and Ionisation Potentials
- VI. Series in Hydrogen Spectrum

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Formula Sheet

I. α -Scattering Experiments

$$r_0 = \frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{E}$$

r_0 = distance of closest approach, Z_1, Z_2 = charge numbers, E = kinetic energy of incident particle

II. Radii of Orbits and Velocities

$$r_n = n^2 r_1, \quad r_1 = 0.529 \text{ \AA (for hydrogen, } Z = 1)$$

r_n = radius of n th Bohr orbit; radius scales as n^2/Z and inversely with orbiting particle's mass

II. Radii of Orbits and Velocities

$$v_n = \frac{e^2}{2\epsilon_0 n h}$$

v_n = orbital speed in n th orbit; independent of the orbiting particle's mass

II. Radii of Orbits and Velocities

$$T = \frac{2\pi r_n}{v_n}$$

T = time period of revolution in the n th orbit

III. Energy of Electron in Different Orbits

$$E_n = -\frac{13.6 Z^2}{n^2} \text{ eV}$$

E_n = total energy in n th orbit; for a Bohr orbit, $KE = -E_n$ and $PE = 2E_n$

IV/V. Transitions, Excitation and Ionisation

$$\Delta E = E_i - E_f = h\nu = \frac{hc}{\lambda}$$

ΔE = photon energy absorbed/emitted; ν = frequency, λ = wavelength

V. Excitation and Ionisation Potentials

$$E_{\text{ionisation}} = -E_1 = 13.6 Z^2 \text{ eV}$$

Ionisation energy = energy to take electron from ground state ($n = 1$) to $n = \infty$; numeri-

cally equal to ionisation potential in volts

VI. Series in Hydrogen Spectrum

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

R = Rydberg constant; n_1 = lower level (series), $n_2 > n_1$ = upper level

VI. Series in Hydrogen Spectrum

$$E_{\text{ionisation}} = Rhc$$

Series-limit (ionisation) energy obtained by putting $n_1 = 1$, $n_2 \rightarrow \infty$

I. α -Scattering Experiments

Question 1

α -particles of 5.0 MeV are being scattered by a thin copper ($Z = 29$) foil. Up to what closest distance can these particles reach the nucleus of copper? If there be protons of the same energy, then?

★ Likely Exam Question

Given: $E = 5.0 \text{ MeV} = 8.0 \times 10^{-13} \text{ J}$; $Z = 29$ (copper); charge on α -particle $= 2e$; charge on proton $= e$; $\frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \text{ N m}^2\text{C}^{-2}$

Formula: $r_0 = \frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{E}$

Solution:

For the α -particle ($Z_1 = 2$, $Z_2 = 29$):

$$\begin{aligned} r_0 &= \frac{9.0 \times 10^9 \times (2 \times 29) \times (1.6 \times 10^{-19})^2}{8.0 \times 10^{-13}} \\ &= \frac{9.0 \times 10^9 \times 58 \times 2.56 \times 10^{-38}}{8.0 \times 10^{-13}} \\ &= 1.67 \times 10^{-14} \text{ m} \end{aligned}$$

For a proton of the same energy ($Z_1 = 1$), the numerator is exactly halved:

$$r'_0 = \frac{9.0 \times 10^9 \times 29 \times 2.56 \times 10^{-38}}{8.0 \times 10^{-13}} = 0.835 \times 10^{-14} \text{ m}$$

Final Answer

$$r_0(\alpha) = 1.67 \times 10^{-14} \text{ m}; \quad r_0(\text{proton}) = 0.835 \times 10^{-14} \text{ m}$$

Common Mistake: Do not forget to use the charge on the α -particle as $2e$ (not e) – this is the most common slip in this type of problem.

Question 2

In a head-on collision between an α -particle and a gold nucleus ($Z = 79$), the distance of closest approach is $4.0 \times 10^{-14} \text{ m}$. Calculate the energy of the particle.

Given: $r_0 = 4.0 \times 10^{-14}$ m; $Z_2 = 79$ (gold); $Z_1 = 2$ (for α -particle); $\frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9$ N m²C⁻²

Formula: $E = \frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{r_0}$

Solution:

$$\begin{aligned} E &= \frac{9.0 \times 10^9 \times (2 \times 79) \times (1.6 \times 10^{-19})^2}{4.0 \times 10^{-14}} \\ &= \frac{9.0 \times 10^9 \times 158 \times 2.56 \times 10^{-38}}{4.0 \times 10^{-14}} \\ &= 9.10 \times 10^{-13} \text{ J} \end{aligned}$$

Converting to MeV:

$$E = \frac{9.10 \times 10^{-13}}{1.6 \times 10^{-13}} \text{ MeV} = 5.69 \text{ MeV}$$

Final Answer

$$E = 5.69 \text{ MeV}$$

II. Radii of Orbits and Velocities in these Orbits

Question 3

The radius of the first orbit of the electron of a hydrogen atom is 5.3×10^{-11} m. What is the radius of its second orbit? (ISC 2006)

★ Likely Exam Question

Given: $r_1 = 5.3 \times 10^{-11}$ m; $n = 2$

Formula: $r_n = n^2 r_1$

Solution:

$$r_2 = (2)^2 \times 5.3 \times 10^{-11} = 4 \times 5.3 \times 10^{-11} = 21.2 \times 10^{-11} \text{ m}$$

Final Answer

$$r_2 = 21.2 \times 10^{-11} \text{ m}$$

Question 4

Show that the velocity of electron in the first orbit of hydrogen atom is $(1/137)$ times the velocity of light.

Given: $n = 1$ (first orbit of hydrogen); $\frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \text{ N m}^2\text{C}^{-2}$; $h = 6.6 \times 10^{-34} \text{ J s}$;
 $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $v_n = \frac{1}{4\pi\epsilon_0} \frac{2\pi e^2}{nh}$

Solution:

For $n = 1$:

$$\begin{aligned} v_1 &= \frac{9.0 \times 10^9 \times 2\pi \times (1.6 \times 10^{-19})^2}{1 \times 6.6 \times 10^{-34}} \\ &= \frac{9.0 \times 10^9 \times 6.283 \times 2.56 \times 10^{-38}}{6.6 \times 10^{-34}} \\ &= 2.19 \times 10^6 \text{ m/s} \end{aligned}$$

Now compare with $c/137$:

$$\frac{c}{137} = \frac{3.0 \times 10^8}{137} = 2.19 \times 10^6 \text{ m/s}$$

Since $v_1 = c/137$, the result is verified.

Final Answer

$$v_1 = 2.19 \times 10^6 \text{ m/s} = \frac{c}{137} \quad (\text{Hence proved})$$

Question 5

A negatively-charged pion (π^-), whose mass is 273 times the mass of electron, moves in a circular orbit around a proton. It is a pi-mesic hydrogen atom. How many revolutions will the pion make during its lifetime of 10^{-8} s in the ground state of the atom?

Given: $m_\pi = 273 m_e$; lifetime $t = 10^{-8}$ s; $n = 1$; $r_1(\text{electron}) = 5.3 \times 10^{-11}$ m; $v_1 = 2.19 \times 10^6$ m/s

Formula: $r_1(\pi) = \frac{r_1(e)}{273}$ (radius $\propto 1/m$, mass only); v_n is independent of mass;
 $T = \frac{2\pi r_1(\pi)}{v_1}$; $N = \frac{t}{T}$

Solution:

The orbital speed v_1 does not depend on the orbiting particle's mass, so it remains 2.19×10^6 m/s. Only the radius shrinks because it is inversely proportional to mass:

$$r_1(\pi) = \frac{5.3 \times 10^{-11}}{273} = 1.94 \times 10^{-13} \text{ m}$$

Time period of revolution:

$$T = \frac{2\pi \times 1.94 \times 10^{-13}}{2.19 \times 10^6} = 5.56 \times 10^{-19} \text{ s}$$

Number of revolutions in the given lifetime:

$$N = \frac{10^{-8}}{5.56 \times 10^{-19}} = 1.8 \times 10^{10}$$

Final Answer

$$N = 1.8 \times 10^{10} \text{ revolutions}$$

Common Mistake: Students often (wrongly) scale the velocity along with the radius. In the Bohr model v_n is completely independent of the orbiting particle's mass – only r_n and E_n depend on it.

III. Energy of Electron in Different Orbits

Question 6

The energy of electron in the first orbit in hydrogen atom is -13.6 eV. What will be the energy in the second orbit?

★ Likely Exam Question

Given: $E_1 = -13.6$ eV; $n = 2$

Formula: $E_n = -\frac{13.6}{n^2}$ eV

Solution:

$$E_2 = -\frac{13.6}{(2)^2} = -\frac{13.6}{4} = -3.4 \text{ eV}$$

Final Answer

$$E_2 = -3.4 \text{ eV}$$

Question 7

Which level of the doubly-ionised lithium (Li^{++}), with $Z = 3$, has the same energy as the ground-state energy of hydrogen atom ($Z = 1$)? Compare the orbital radii of the two levels.

Given: $Z_{\text{Li}} = 3$; $Z_{\text{H}} = 1$; $E_1(\text{H}) = -13.6$ eV

Formula: $E_n = -\frac{13.6 Z^2}{n^2}$ eV; $r_n = \frac{0.529 n^2}{Z}$ Å

Solution:

Equate the energy of Li^{++} level n to the hydrogen ground-state energy:

$$-\frac{13.6 \times 3^2}{n^2} = -13.6 \Rightarrow \frac{9}{n^2} = 1 \Rightarrow n^2 = 9 \Rightarrow n = 3$$

Ratio of orbital radii:

$$\frac{r_3(\text{Li}^{++})}{r_1(\text{H})} = \frac{n^2/Z_{\text{Li}}}{1^2/Z_{\text{H}}} = \frac{9/3}{1} = 3$$

Final Answer

$$n = 3 \text{ level of Li}^{++}; \quad \frac{r_3(\text{Li}^{++})}{r_1(\text{H})} = 3$$

IV. Energy Emitted/Absorbed in Transitions

Question 8

The electron in a given Bohr orbit has a total energy of -1.5 eV. Calculate its (i) kinetic energy (ii) potential energy and (iii) wavelength of light emitted, when electron makes a transition to the ground state. Ground state energy is -13.6 eV.

Given: $E = -1.5$ eV; $E_1 = -13.6$ eV

Formula: $KE = -E$; $PE = 2E$; $\lambda = \frac{hc}{\Delta E}$

Solution:

(i) Kinetic energy:

$$KE = -E = -(-1.5) = 1.5 \text{ eV}$$

(ii) Potential energy:

$$PE = 2E = 2 \times (-1.5) = -3.0 \text{ eV}$$

(iii) Energy released on transition to ground state:

$$\Delta E = (-1.5) - (-13.6) = 12.1 \text{ eV} = 12.1 \times 1.6 \times 10^{-19} \text{ J} = 1.936 \times 10^{-18} \text{ J}$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{1.936 \times 10^{-18}} = 1022.7 \times 10^{-10} \text{ m} = 1022.7 \text{ \AA}$$

Final Answer

(i) $KE = 1.5$ eV (ii) $PE = -3.0$ eV (iii) $\lambda = 1022.7 \text{ \AA}$

Common Mistake: Total energy $E = -KE = PE/2$ holds only in the Bohr model; students often confuse the sign of PE.

Question 9

The energy of an electron in the n th orbit of hydrogen atom is given by $E_n = -(13.6/n^2)$ eV. Find energy required to take the electron from the ground state to the (i) first excited state (ii) second excited state.

★ Likely Exam Question

Given: $E_1 = -13.6$ eV; first excited state $n = 2$; second excited state $n = 3$

Formula: $\Delta E = E_n - E_1$

Solution:(i) To $n = 2$:

$$\Delta E = -\frac{13.6}{4} - (-13.6) = -3.4 + 13.6 = 10.2 \text{ eV}$$

(ii) To $n = 3$:

$$\Delta E = -\frac{13.6}{9} - (-13.6) = -1.51 + 13.6 = 12.09 \text{ eV}$$

Final Answer

(i) 10.2 eV (ii) 12.09 eV

Question 10

Calculate the frequency of the photon, which can excite the electron to -3.4 eV from -13.6 eV .

Given: $E_i = -13.6 \text{ eV}$; $E_f = -3.4 \text{ eV}$; Hint: $\Delta E = h\nu$

Formula: $\nu = \frac{\Delta E}{h}$

Solution:

$$\Delta E = -3.4 - (-13.6) = 10.2 \text{ eV} = 10.2 \times 1.6 \times 10^{-19} = 1.632 \times 10^{-18} \text{ J}$$

$$\nu = \frac{1.632 \times 10^{-18}}{6.6 \times 10^{-34}} = 2.47 \times 10^{15} \text{ Hz}$$

Final Answer

$$\nu = 2.47 \times 10^{15} \text{ Hz}$$

Question 11

The energy of an electron in the n th state of hydrogen is given by $E_n = -(13.6/n^2) \text{ eV}$. Calculate the energy and frequency of a photon radiated when the electron de-excites from the first excited state to the ground state. To which spectral series does this photon belong?

Given: first excited state $n = 2$; ground state $n = 1$

Formula: $\Delta E = E_1 - E_2$; $\nu = \Delta E/h$

Solution:

$$E_1 = -13.6 \text{ eV}, \quad E_2 = -3.4 \text{ eV}$$

$$\Delta E = |-13.6 - (-3.4)| = 10.2 \text{ eV} = 1.632 \times 10^{-18} \text{ J}$$

$$\nu = \frac{1.632 \times 10^{-18}}{6.6 \times 10^{-34}} = 2.47 \times 10^{15} \text{ Hz}$$

Since the electron lands on $n = 1$, the photon belongs to the **Lyman series**.

Final Answer

$$\Delta E = 10.2 \text{ eV}, \nu = 2.47 \times 10^{15} \text{ Hz, Lyman series}$$

Question 12

The energy of the lowest level of hydrogen atom is -13.6 eV . Calculate the energy, frequency and wavelength of the photon emitted by transition from $n = 4$ to $n = 2$.

★ Likely Exam Question

Given: $n_i = 4; n_f = 2; E_1 = -13.6 \text{ eV}$

Formula: $E_n = -\frac{13.6}{n^2} \text{ eV}; \quad \nu = \frac{\Delta E}{h}; \quad \lambda = \frac{c}{\nu}$

Solution:

$$E_4 = -\frac{13.6}{16} = -0.85 \text{ eV}, \quad E_2 = -\frac{13.6}{4} = -3.4 \text{ eV}$$

$$\Delta E = |-0.85 - (-3.4)| = 2.55 \text{ eV} = 2.55 \times 1.6 \times 10^{-19} = 4.08 \times 10^{-19} \text{ J}$$

$$\nu = \frac{4.08 \times 10^{-19}}{6.6 \times 10^{-34}} = 6.18 \times 10^{14} \text{ Hz}$$

$$\lambda = \frac{3.0 \times 10^8}{6.18 \times 10^{14}} = 4.853 \times 10^{-7} \text{ m} = 4853 \text{ Å}$$

Final Answer

$$\Delta E = 2.55 \text{ eV}, \nu = 6.18 \times 10^{14} \text{ Hz}, \lambda = 4853 \text{ Å}$$

Question 13

The energy of electron orbiting in the n th orbit of hydrogen atom is given by $E_n = -(13.6/n^2) \text{ eV}$. Calculate: (i) energy liberated when the electron jumps from the second to the first orbit (ii) wavelength of the light emitted in this transition (iii) the ionisation energy of the atom.

★ Likely Exam Question

Given: $n_i = 2, n_f = 1; E_1 = -13.6 \text{ eV}, E_2 = -3.4 \text{ eV}$

Formula: $\Delta E = E_1 - E_2; \quad \lambda = \frac{hc}{\Delta E}; \quad E_{\text{ionisation}} = -E_1$

Solution:

(i)

$$\Delta E = |-13.6 - (-3.4)| = 10.2 \text{ eV}$$

(ii)

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{10.2 \times 1.6 \times 10^{-19}} = \frac{1.98 \times 10^{-25}}{1.632 \times 10^{-18}} = 1213 \times 10^{-10} \text{ m} = 1213 \text{ \AA}$$

(iii)

$$E_{\text{ionisation}} = -E_1 = 13.6 \text{ eV}$$

Final Answer

(i) 10.2 eV (ii) 1213 Å (iii) 13.6 eV

Question 14

The energies of two energy-levels of hydrogen atom are -3.4 eV and -1.5 eV . If electron transition occurs from the higher to the lower level, how much energy will be emitted? What will be the wavelength of the emitted radiation?

Given: $E_{\text{higher}} = -1.5 \text{ eV}$; $E_{\text{lower}} = -3.4 \text{ eV}$

Formula: $\Delta E = E_{\text{higher}} - E_{\text{lower}}$; $\lambda = \frac{hc}{\Delta E}$

Solution:

$$\Delta E = -1.5 - (-3.4) = 1.9 \text{ eV} = 1.9 \times 1.6 \times 10^{-19} = 3.04 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{3.04 \times 10^{-19}} = 6.513 \times 10^{-7} \text{ m} = 6513 \text{ \AA}$$

Final Answer $\Delta E = 1.9 \text{ eV}$, $\lambda = 6513 \text{ \AA}$

Common Mistake: The less negative energy is the higher energy state (-1.5 eV is higher than -3.4 eV) – do not simply subtract the smaller magnitude from the larger one without checking signs.

Question 15

The energy of hydrogen atom in its ground state is -13.6 eV . The atom is given an energy of 10.2 eV . In which state will it reach? What will be the wavelength of the absorbed photon in this process?

Given: $E_1 = -13.6 \text{ eV}$; energy absorbed = 10.2 eV

Formula: $E_n = E_1 + \Delta E$; $\lambda = \frac{hc}{\Delta E}$

Solution:

$$E_n = -13.6 + 10.2 = -3.4 \text{ eV}$$

Since $E_2 = -13.6/4 = -3.4 \text{ eV}$, the atom reaches the state $n = 2$.

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{10.2 \times 1.6 \times 10^{-19}} = 1213 \times 10^{-10} \text{ m} = 1213 \text{ \AA}$$

Final Answer

$n = 2$ (first excited state); $\lambda = 1213 \text{ \AA}$

V. Excitation and Ionisation Potentials

Question 16

The wavelength of the first spectral line of sodium is $5896 \text{ \AA}$. What is the first excitation potential of sodium atom?

Given: $\lambda = 5896 \text{ \AA} = 5896 \times 10^{-10} \text{ m}$

Formula: $\Delta E = \frac{hc}{\lambda}$; excitation potential (V) = ΔE (in eV)

Solution:

$$\Delta E = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{5896 \times 10^{-10}} = 3.358 \times 10^{-19} \text{ J}$$

$$\Delta E = \frac{3.358 \times 10^{-19}}{1.6 \times 10^{-19}} = 2.1 \text{ eV}$$

Final Answer

First excitation potential = 2.1 V

Question 17

The ionisation-potential of hydrogen atom is 13.6 V. Find out: (i) wavelength of H_β line of Balmer series (ii) shortest wavelength of Lyman series.

★ Likely Exam Question

Given: ionisation energy = 13.6 eV; H_β : $n = 4 \rightarrow n = 2$

Formula: $\Delta E = E_{n_f} - E_{n_i}$; $\lambda = \frac{hc}{\Delta E}$

Solution:

(i) For H_β ($n = 4 \rightarrow 2$):

$$E_4 = -0.85 \text{ eV}, \quad E_2 = -3.4 \text{ eV}, \quad \Delta E = 2.55 \text{ eV}$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{2.55 \times 1.6 \times 10^{-19}} = 4853 \times 10^{-10} \text{ m} = 4853 \text{ \AA}$$

(ii) Shortest wavelength of Lyman series corresponds to $n = \infty \rightarrow n = 1$, i.e. $\Delta E = 13.6 \text{ eV}$:

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{13.6 \times 1.6 \times 10^{-19}} = 910 \times 10^{-10} \text{ m} = 910 \text{ \AA}$$

Final Answer

(i) $\lambda(H_\beta) = 4853 \text{ \AA}$ (ii) $\lambda_{\text{shortest, Lyman}} = 910 \text{ \AA}$

Question 18

Calculate the wavelength of the H_α line of hydrogen spectrum. The energy levels of hydrogen atom are given by the expression $E_n = -(13.6/n^2) \text{ eV}$.

Given: H_α line: $n = 3 \rightarrow n = 2$

Formula: $\Delta E = E_2 - E_3; \quad \lambda = \frac{hc}{\Delta E}$

Solution:

$$E_3 = -\frac{13.6}{9} = -1.51 \text{ eV}, \quad E_2 = -\frac{13.6}{4} = -3.4 \text{ eV}$$

$$\Delta E = |-3.4 - (-1.51)| \approx 1.89 \text{ eV} = 1.89 \times 1.6 \times 10^{-19} = 3.024 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{3.024 \times 10^{-19}} = 6547.6 \times 10^{-10} \text{ m} = 6547.6 \text{ \AA}$$

Final Answer

$\lambda(H_\alpha) = 6547.6 \text{ \AA}$

Question 19

The first excitation potential of mercury is 4.86 V. A mercury atom, after absorbing a photon, reaches its first excited state. Find the wavelength of the absorbed photon.

Given: excitation potential = 4.86 V $\Rightarrow \Delta E = 4.86 \text{ eV}$

Formula: $\lambda = \frac{hc}{\Delta E}$

Solution:

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{4.86 \times 1.6 \times 10^{-19}} = \frac{1.98 \times 10^{-25}}{7.776 \times 10^{-19}} = 2546 \times 10^{-10} \text{ m} = 2546 \text{ \AA}$$

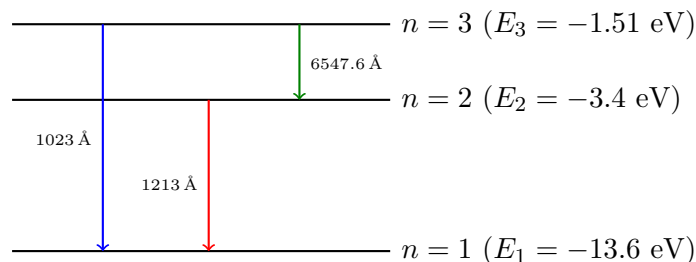
Final Answer

$\lambda = 2546 \text{ \AA}$

Question 20

The first and second excitation potentials of hydrogen atom are respectively 10.2 V and 12.09 V. Draw, on this basis, the energy-level diagram of the atom and show all possible emission transitions in it and mark their corresponding wavelengths.

★ Likely Exam Question



Energy-level diagram of hydrogen showing the three possible emission transitions ($n=2 \rightarrow 1$, $n=3 \rightarrow 1$, $n=3 \rightarrow 2$)

Given: first excitation potential = 10.2 V $\Rightarrow E_2 = -3.4$ eV; second excitation potential = 12.09 V $\Rightarrow E_3 = -1.51$ eV; $E_1 = -13.6$ eV

Formula: $\Delta E = E_{n_f} - E_{n_i}$; $\lambda = \frac{hc}{\Delta E}$

Solution:

Three transitions are possible: $2 \rightarrow 1$, $3 \rightarrow 1$, $3 \rightarrow 2$.

$$\Delta E(2 \rightarrow 1) = 10.2 \text{ eV} \Rightarrow \lambda = \frac{1.98 \times 10^{-25}}{10.2 \times 1.6 \times 10^{-19}} = 1213 \text{ Å}$$

$$\Delta E(3 \rightarrow 1) = 12.09 \text{ eV} \Rightarrow \lambda = \frac{1.98 \times 10^{-25}}{12.09 \times 1.6 \times 10^{-19}} = 1023 \text{ Å}$$

$$\Delta E(3 \rightarrow 2) = 12.09 - 10.2 = 1.89 \text{ eV} \Rightarrow \lambda = \frac{1.98 \times 10^{-25}}{1.89 \times 1.6 \times 10^{-19}} = 6547.6 \text{ Å}$$

Final Answer

$$\lambda(2 \rightarrow 1) = 1213 \text{ Å}, \lambda(3 \rightarrow 1) = 1023 \text{ Å}, \lambda(3 \rightarrow 2) = 6547.6 \text{ Å}$$

Question 21

The ionisation potential of hydrogen atom is 13.6 V. In the lowest energy-level, this atom is ionised by absorbing a photon of 800 Å. What will be the kinetic energy of the released electron?

Given: ionisation energy = 13.6 eV; $\lambda_{\text{photon}} = 800 \text{ Å} = 800 \times 10^{-10} \text{ m}$

Formula: $E_{\text{photon}} = \frac{hc}{\lambda}$; $KE_{\text{electron}} = E_{\text{photon}} - E_{\text{ionisation}}$

Solution:

$$E_{\text{photon}} = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{800 \times 10^{-10}} = 2.475 \times 10^{-18} \text{ J} = \frac{2.475 \times 10^{-18}}{1.6 \times 10^{-19}} = 15.47 \text{ eV}$$

$$KE = 15.47 - 13.6 = 1.87 \text{ eV}$$

Final Answer

$$KE = 1.87 \text{ eV}$$

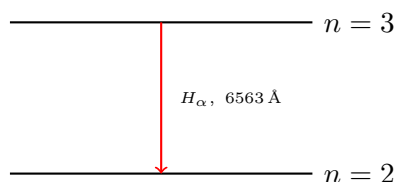
Common Mistake: The photon energy must exceed the ionisation energy; the *excess* energy (not the full photon energy) becomes the electron's kinetic energy.

VI. Series in Hydrogen Spectrum

Question 22

Calculate the wavelength of the first line of the Balmer series of hydrogen. Indicate the transition on an energy-level diagram. (i) What is this line called? (ii) Does it lie in the visible region of the electromagnetic spectrum?

★ Likely Exam Question



First line of the Balmer series: transition $n = 3 \rightarrow n = 2$, the H_α line

Given: $n_i = 3$, $n_f = 2$; $R = 1.097 \times 10^7 \text{ m}^{-1}$

Formula: $\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$

Solution:

$$\frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{4} - \frac{1}{9} \right) = 1.097 \times 10^7 \times \frac{5}{36} = 1.524 \times 10^6 \text{ m}^{-1}$$

$$\lambda = \frac{1}{1.524 \times 10^6} = 6563 \times 10^{-10} \text{ m} = 6563 \text{ Å}$$

(i) This line is called the H_α **line**. (ii) **Yes**, 6563 Å lies in the visible (red) region of the electromagnetic spectrum.

Final Answer

$\lambda = 6563 \text{ Å}$; (i) H_α line; (ii) Yes, lies in the visible region

Question 23

The wavelength of the first line of Balmer series is 6563 Å. Calculate (i) Rydberg's constant, (ii) ionisation potential of hydrogen atom.

★ Likely Exam Question

Given: $\lambda = 6563 \text{ \AA} = 6563 \times 10^{-10} \text{ m}$ (first Balmer line, $n = 3 \rightarrow n = 2$)

Formula: $\frac{1}{\lambda} = R \left(\frac{1}{4} - \frac{1}{9} \right); \quad E_{\text{ionisation}} = Rhc$

Solution:

(i)

$$R = \frac{36}{5\lambda} = \frac{36}{5 \times 6563 \times 10^{-10}} = 1.097 \times 10^7 \text{ m}^{-1}$$

(ii)

$$E_{\text{ionisation}} = Rhc = 1.097 \times 10^7 \times 6.6 \times 10^{-34} \times 3.0 \times 10^8 = 2.172 \times 10^{-18} \text{ J}$$

$$E_{\text{ionisation}} = \frac{2.172 \times 10^{-18}}{1.6 \times 10^{-19}} = 13.6 \text{ eV}$$

Final Answer

(i) $R = 1.097 \times 10^7 \text{ m}^{-1}$ (ii) Ionisation potential = 13.6 V

Question 24

The energy transition in hydrogen atom takes place from energy level $n = 8$ to $n = 1$. Calculate the wavelength of the emitted photon. To which spectral series will this photon belong? To which part of the electromagnetic spectrum will this photon belong? ($R = 1.09 \times 10^7$ per metre.)

★ Likely Exam Question

Given: $n_i = 8, n_f = 1; R = 1.09 \times 10^7 \text{ m}^{-1}$

Formula: $\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$

Solution:

$$\frac{1}{\lambda} = 1.09 \times 10^7 \left(1 - \frac{1}{64} \right) = 1.09 \times 10^7 \times \frac{63}{64} = 1.073 \times 10^7 \text{ m}^{-1}$$

$$\lambda = \frac{1}{1.073 \times 10^7} = 932 \times 10^{-10} \text{ m} = 932 \text{ \AA}$$

Since the electron lands on $n = 1$, this photon belongs to the **Lyman series**, which lies in the **ultraviolet** region of the electromagnetic spectrum.

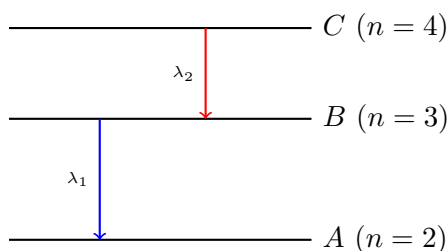
Final Answer

$\lambda = 932 \text{ \AA}$; Lyman series; ultraviolet region

Question 25

As per the given figure, A , B and C are the first, second and third excited states of a hydrogen atom respectively. If the ratio of the two wavelengths (i.e. λ_1/λ_2) is $7/4n$, find the value of n .

★ Likely Exam Question



Energy-level diagram: $A(n=2)$, $B(n=3)$, $C(n=4)$; λ_1 is the transition $B \rightarrow A$, λ_2 is the transition $C \rightarrow B$

Given: $A : n = 2$ (first excited state); $B : n = 3$ (second excited state); $C : n = 4$ (third excited state); λ_1 : transition $B \rightarrow A$; λ_2 : transition $C \rightarrow B$

Formula: $E_n = -\frac{13.6}{n^2} \text{ eV}$; $\lambda \propto \frac{1}{\Delta E}$

Solution:

$$E_A = -\frac{13.6}{4} = -3.4 \text{ eV}, \quad E_B = -\frac{13.6}{9} = -1.511 \text{ eV}, \quad E_C = -\frac{13.6}{16} = -0.85 \text{ eV}$$

$$\Delta E_1(B \rightarrow A) = |-3.4 - (-1.511)| = 1.889 \text{ eV}$$

$$\Delta E_2(C \rightarrow B) = |-1.511 - (-0.85)| = 0.661 \text{ eV}$$

Since $\lambda \propto 1/\Delta E$:

$$\frac{\lambda_1}{\lambda_2} = \frac{\Delta E_2}{\Delta E_1} = \frac{0.661}{1.889} = 0.35 = \frac{7}{20}$$

Comparing with the given ratio $\frac{7}{4n}$:

$$\frac{7}{4n} = \frac{7}{20} \Rightarrow 4n = 20 \Rightarrow n = 5$$

Final Answer

$$n = 5$$

Question 26

Calculate the wavelengths of the third and the fifth spectral lines of Paschen series in hydrogen spectrum when $E = -(2.18 \times 10^{-18})/n^2$ joule.

Given: Paschen series: transitions terminate at $n_f = 3$; third line: $n_i = 6$; fifth line:

$$n_i = 8; E_n = -\frac{2.18 \times 10^{-18}}{n^2} \text{ J}$$

Formula: $\Delta E = E_{n_f} - E_{n_i}; \quad \lambda = \frac{hc}{\Delta E}$

Solution:

Third line ($n = 6 \rightarrow n = 3$):

$$E_3 = -\frac{2.18 \times 10^{-18}}{9} = -2.422 \times 10^{-19} \text{ J}, \quad E_6 = -\frac{2.18 \times 10^{-18}}{36} = -6.056 \times 10^{-20} \text{ J}$$

$$\Delta E = |-2.422 \times 10^{-19} - (-6.056 \times 10^{-20})| = 1.817 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{1.817 \times 10^{-19}} = 10899 \times 10^{-10} \text{ m} = 10899 \text{ Å}$$

Fifth line ($n = 8 \rightarrow n = 3$):

$$E_8 = -\frac{2.18 \times 10^{-18}}{64} = -3.406 \times 10^{-20} \text{ J}$$

$$\Delta E = |-2.422 \times 10^{-19} - (-3.406 \times 10^{-20})| = 2.082 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{2.082 \times 10^{-19}} = 9512 \times 10^{-10} \text{ m} = 9512 \text{ Å}$$

Final Answer

Third line: $\lambda = 10899 \text{ Å}$; Fifth line: $\lambda = 9512 \text{ Å}$

Common Mistake: In the Paschen series all lines terminate at $n_f = 3$, not $n_f = 1$; the third line is the transition $n = 6 \rightarrow 3$ (first line is $4 \rightarrow 3$, second is $5 \rightarrow 3$, third is $6 \rightarrow 3$), not $n = 3 \rightarrow 3$ or similar miscounting.

Question 27

The wavelength of a yellow line of sodium is 5896 Å . Give its wave number.

Given: $\lambda = 5896 \text{ Å} = 5896 \times 10^{-10} \text{ m} = 5896 \times 10^{-8} \text{ cm}$

Formula: $\bar{\nu} = \frac{1}{\lambda}$

Solution:

$$\bar{\nu} = \frac{1}{5896 \times 10^{-8} \text{ cm}} = 16961 \text{ cm}^{-1}$$

Final Answer

$\bar{\nu} = 16961 \text{ per cm}$

Question 28

The wavelength of the first line of Balmer series is 6563 Å. Find the wavelength of the first line of the Lyman series.

★ Likely Exam Question

Given: $\lambda_{\text{Balmer},1} = 6563 \text{ Å}$ ($n = 3 \rightarrow 2$); Lyman first line: $n = 2 \rightarrow 1$

Formula: $\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$

Solution:

For the Balmer first line: $\frac{1}{\lambda_B} = R \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{5R}{36}$

For the Lyman first line: $\frac{1}{\lambda_L} = R \left(1 - \frac{1}{4} \right) = \frac{3R}{4}$

Taking the ratio (so that R cancels):

$$\frac{\lambda_L}{\lambda_B} = \frac{5R/36}{3R/4} = \frac{5}{36} \times \frac{4}{3} = \frac{5}{27}$$

$$\lambda_L = 6563 \times \frac{5}{27} = 1215 \text{ Å}$$

Final Answer

$$\lambda(\text{Lyman, first line}) = 1215 \text{ Å}$$

Question 29

The wavelength of the first member of the Balmer series is 6563 Å. Find the wavelengths of the second and the last member of the series.

Given: $\lambda_1 = 6563 \text{ Å}$ (first member, $n = 3 \rightarrow 2$); second member: $n = 4 \rightarrow 2$; last member: $n = \infty \rightarrow 2$

Formula: $\frac{1}{\lambda} = R \left(\frac{1}{4} - \frac{1}{n_i^2} \right)$

Solution:

First member coefficient: $\frac{1}{\lambda_1} \propto \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{5}{36}$

Second member ($n = 4 \rightarrow 2$): $\frac{1}{\lambda_2} \propto \left(\frac{1}{4} - \frac{1}{16} \right) = \frac{3}{16}$

$$\lambda_2 = \lambda_1 \times \frac{5/36}{3/16} = 6563 \times \frac{80}{108} = 4861 \text{ Å}$$

Last member ($n = \infty \rightarrow 2$, series limit): $\frac{1}{\lambda_{\text{last}}} \propto \frac{1}{4}$

$$\lambda_{\text{last}} = \lambda_1 \times \frac{5/36}{1/4} = 6563 \times \frac{20}{36} = 3646 \text{ \AA}$$

Final Answer

Second member: $\lambda_2 = 4861 \text{ \AA}$; Last member: $\lambda_{\text{last}} = 3646 \text{ \AA}$

Question 30

The ionisation energy of hydrogen atom is 13.6 eV. Calculate Rydberg's constant for hydrogen.

Given: $E_{\text{ionisation}} = 13.6 \text{ eV}$; Hint: $Rhc = 13.6 \text{ eV}$

Formula: $R = \frac{E_{\text{ionisation}}}{hc}$

Solution:

$$E_{\text{ionisation}} = 13.6 \times 1.6 \times 10^{-19} = 2.176 \times 10^{-18} \text{ J}$$

$$R = \frac{2.176 \times 10^{-18}}{6.6 \times 10^{-34} \times 3.0 \times 10^8} = \frac{2.176 \times 10^{-18}}{1.98 \times 10^{-25}} = 1.0956 \times 10^7 \text{ m}^{-1}$$

Final Answer

$$R = 1.0956 \times 10^7 \text{ m}^{-1}$$